Hochschule Karlsruhe
University of
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Fakultät für
Maschinenbau und
Mechatronik

LaTeX Template For Dissertations, Master Theses, Bachelor Theses, Student Projects, And Other Reports at the Institute of Energy Efficient Mobility

Forschungs- und Entwicklungsprojekt 1 (Master)

am Institut für Energieeffiziente Mobilität
Fakultät für Maschinenbau und Mechatronik
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Fassung vom:1. März 2024

Erklärung

Hiermit versichere ich, dass ich die vorliegende Arbeit mit dem Titel

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selbstständig angefertigt, alle benutzten Hilfsmittel vollständig und genau angegeben und alles einzeln kenntlich gemacht zu haben, was aus Arbeiten anderer unverändert oder mit Abänderungen entnommen wurde.

Karlsruhe, den 1. März 2024

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Zusammenfassung

A short summary of the work should be given. This should not exceed one page. The relevant aspects that should be addressed are:

1. What is the problem/motivation addressed in this work?
2. Why is the problem relevant?
3. What is the current state of the art / why has the problem not been addressed so far?
4. What is your solution?
5. What are the main results and contributions of your work that addressed the original problem

To get an orientation how an abstract can look like, you can take a look into published scientific papers or dissertations.

Abstract

This is a translation of the abstract into another language. In case your work is written in German, this page is usually in English and vice versa.

Inhaltsverzeichnis

1 Introduction

This template provides a LaTeX environment that can be used for the preparation of project papers, theses and dissertations.

This document only provides the basic structure. For content suggestions, please refer to the following documents:

- *Hinweise Abschlussarbeiten doppelseitig 2019-09-23* by Prof. Dr.-Ing. Michael Arnemann. This document is a university-wide guideline[1]. **Attention:** The formal structure of this template (e.g. page formats, cover page, etc.) already largely follows Mr. Arnemann's guidelines and is adapted to the requirements of the IEEM. You therefore do not need to change the report format. However, Mr. Arnemann's guidelines provide valuable advice on content such as the presentation of figures, tables, equations, etc. and their labels. In addition, advice is given on citation, the structure and content of chapters and other aspects of the work.
- *Leitfaden: Wissenschaftliches Arbeiten (Schwerpunkt: Ablauf der Arbeit, Schreiben des Berichts, Halten einer Präsentation) by Florian Sommer*. This document is related to the IEEM and gives advice on how to organize a project and final thesis, how to prepare the written report and how to give the final presentation. It primarily provides tips on content, such as the recommended length of the thesis, the best way to start, what content should be included in the thesis or presentation, etc. In particular, tips are given on the structure and implementation of individual report chapters and presentation slides, as well as an overview of correct citation and the use of illustrations.

We recommend that you read both documents thoroughly and consider them in your work. Experience has shown that this leads to better reports and final presentations and subsequently to better grades for your work.

[1] The guideline is not a specification that must be followed exactly in every single point. However, it offers useful tips for the implementation of a project or final thesis, which makes suggestions for the design and content of final theses.

1.1 Comments on the Format

This work is double-sided. On even pages, the current chapter is automatically displayed in the header. On odd-numbered pages, the current section is displayed in the header (note: only the section, but not the subsection or subsubsection). Chapter beginnings are excluded from this, as these pages do not have a header.

Page numbers are automatically assigned and displayed in the footer. No page numbers are displayed on empty pages (e.g., at the end of a chapter).

Footnotes[2] can be automatically inserted in the footer of the page on which you have specified the footer.

Illustrations, tables, and other aspects are presented below as examples.

1.1.1 Figures

In Abbildung 1.1, the example of an automotive E/E architecture is illustrated.

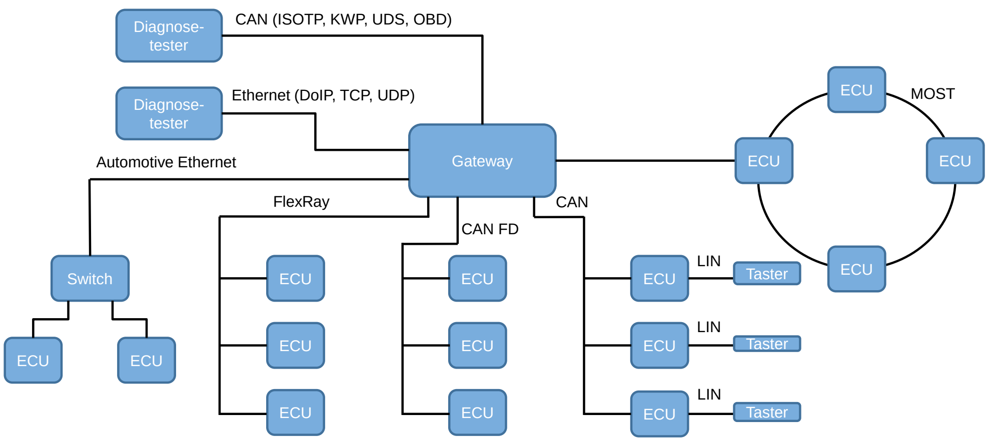

Abbildung 1.1: Overview of an automotive E/E architecture and its individual subsystems. The Electronic Control Units (ECUs) are connected by five bus systems. Furthermore, a diagnostic tester is connected by a Controller Area Network (CAN) bus and Ethernet.

The caption of a figure should consist of two sentences at least, since you need to describe what the figure is showing. **Attention:** You always have to pay attention to copyrights when you use images!!!

[2]This is a footnote.

1.1.2 Tables

In Tabelle 1.1, the example of a table is shown.

Tabelle 1.1: Table in which the \toprule, \toprule, and \toprule commands are used without vertical separators. This table uses the tabularx package. The table width is fixed. There are multiple options to create tables.

Country	**City**	**Students**
France	Paris	500
Germany	Karlsruhe	450

In Tabelle 1.2, another table is shown.

Tabelle 1.2: Table in which the \hline command and the vertical separators are used to create the cells. There are multiple options to create tables.

1	2	3
4	5	6
7	8	9

1.1.3 Algorithms

In Algorithmus 1.1, the example of a pseudocode algorithm is shown.

```
function TestFunction(i ∈ I)                              ▷ Function Header
    init m ← 0                                            ▷ Initializing values
    init M ← {1,2,3,4,5}

    if (i > 0) ∪ (i ≤ 5) then                             ▷ If-Else
        m ← 5
    else if (i > 5) ∪ (i < 10) then
        m ← 10
    else
        m ← 25
    end if

    while i > 0 do                                        ▷ While-Loop
        m ← m + 1
    end while

    for all n ∈ M do                                      ▷ Forall-Loop
        output n                                          ▷ Output (Print) of a number
    end for

    for n = 1,...,10 do                                   ▷ For-Loop
        output n
    end for

    CalculateValue(m, i)                                  ▷ Function call with parameters
    return m      ▷ Returning the number m, um zu prüfen, ob die breite passt
end function
```

Algorithmus 1.1: Exemplary algorithm to demonstrate pseudocode in this thesis.

Pesudocode algorithms in this document use the LaTeX package *Algpseudocodex*. Pseudocode is typically a mix of program code, mathematical or logical symbols and operations, and textual descriptions. The purpose of pseudocode is the description of an algorithm, function, or procedure in an easy way. The focus is on readability and understandability. The formal precision of the expressions plays a minor role.

1.1.4 Code Listings

In Listing 1.1, the example of program code. The default option is C/C++.

```

void function(int limit)
{
    for (int i = 0; i < limit; i++)  // This is a comment!
    {
        if (i % 15 == 0)
        {
            printf("Test");
        }
        else if (i % 3 == 0)
        {
            printf("String");
        }
        else if (i % 5 == 0)
        {
            int var = 0123456789;
            printf("Print");
        }
        else
        {
            printf(i);
        }
    }
}
```

Listing 1.1: Code example.

We recommend you to place program code in annex chapters, but not in normal text chapters. In the text, it is more reasonable to illustrate the behavior of your programm in flow charts, state machines, pseudocode, etc., since these are easier to read and understand.

1.1.5 Further Remarks

The pages and captions of all figures, tables, algorithms, and code listings are automatically referenced in the *List of Figures, Tables*, etc. at the end of your document.

2 Background

3 Concept

4 Implementation

5 Evaluation

6 Conclusion

A Headline of Appendix A

In principle, the appendix contains things that are too large or too extensive to be mentioned directly in the text:

- Large tables
- Program code
- Large figures
- Measurement data
- ...

Furthermore, the appendix can be used to give detailed explanations of aspects that are to long to cover them directly in background or state of the art chapters. The appendix can also be structured in chapters (Anhang A, Anhang B, ...)

B Headline of Appendix B

Abbildungsverzeichnis

Tabellenverzeichnis

Algorithmenverzeichnis

Programmcodeverzeichnis

Abkürzungsverzeichnis

CAN	Controller Area Network
ECU	Electronic Control Unit

Symbolverzeichnis

$I, Item$	Definition of a system model item
$Items$	Set of all system model items
$M_{Adjacent}$	Adjacency matrix of the system model, which defines connected items

Literatur

[1] Frank Kargl, Natasa Trkulja, Artur Hermann, Florian Sommer, Anderson Ramon Ferraz de Lucena, Alexander Kiening und Sergej Japs. „Securing Cooperative Intersection Management through Subjective Trust Networks“. In: *2023 IEEE 97th Vehicular Technology Conference (VTC2023-Spring), Florence, Italy*. IEEE, 2023, S. 1–7. ISBN: 979-8-3503-1114-3. DOI: 10.1109/VTC2023-Spring57618.2023.10200789.

[2] Patrick Rebling, Anna-Lena Marmein, Magdalena Kugler, Yannick Rauch und Florian Sommer. „Development and Testing of Driver Assistance Systems with Unity“. In: *Reports on Energy Efficient Mobility* 1 (2020), S. 51–55. DOI: 10.5281/zenodo.4569339.

[3] Florian Sommer und Jürgen Dürrwang. *IEEM-HsKA/AAD: Automotive Attack Database (AAD)*. 2019. URL: https://github.com/IEEM-HsKA/AAD (besucht am 3. Dez. 2023).